

Open set classification: application to seismic data

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Abstract

Open set classification deals with building classification models that are able to handle data points that arrive from un-known classes. Unlike traditional classifiers, which assume that only known classes will appear in the test set, open set classifiers are designed to handle the presence of unknown classes. One application field in which open set classification is needed, but still not developed, is geophysical classification tasks, for example, classification of seismic signals. In this work we first explore closed-set classification of quarry blasts, in order to extend the developed method to an open-set model.

Introduction

Quarry explosion signals are collected as time series. The main challenge in time series classification is how to represent the data to reach a sufficient level of classification by few shots. We explore and compare several representation techniques, some are time-frequency based and other state-of-the-art convolution methods. Closed-set classifiers are then used to evaluate the different representations of the seismic signals. The seismic dataset includes several hundreds of quarry blast explosions from Israel and Jordan (see the map in Figure 1). The data was supplied by Soreq Nuclear Research Center, who collaborate on this project.

Since explosion signals can arrive from other sources and locations that did not appear in the training-set, we outline a framework to deal with an open set classification, with the ability to reject out-of-distribution samples. In machine learning, accurately distinguishing between in-distribution and out-of-distribution data is essential for building robust models, especially in critical applications where misclassifications can have significant consequences. To do so, we outline the main steps of a method that addresses the challenges for the seismic signal classification task. The proposed method may be applied to other data types.

Experimental results focus on selecting the best representation method in the setting of closed-set classification. We found the time-frequency sonogram representation yields the best results and performs well in the setting of few- or one-shot training set.

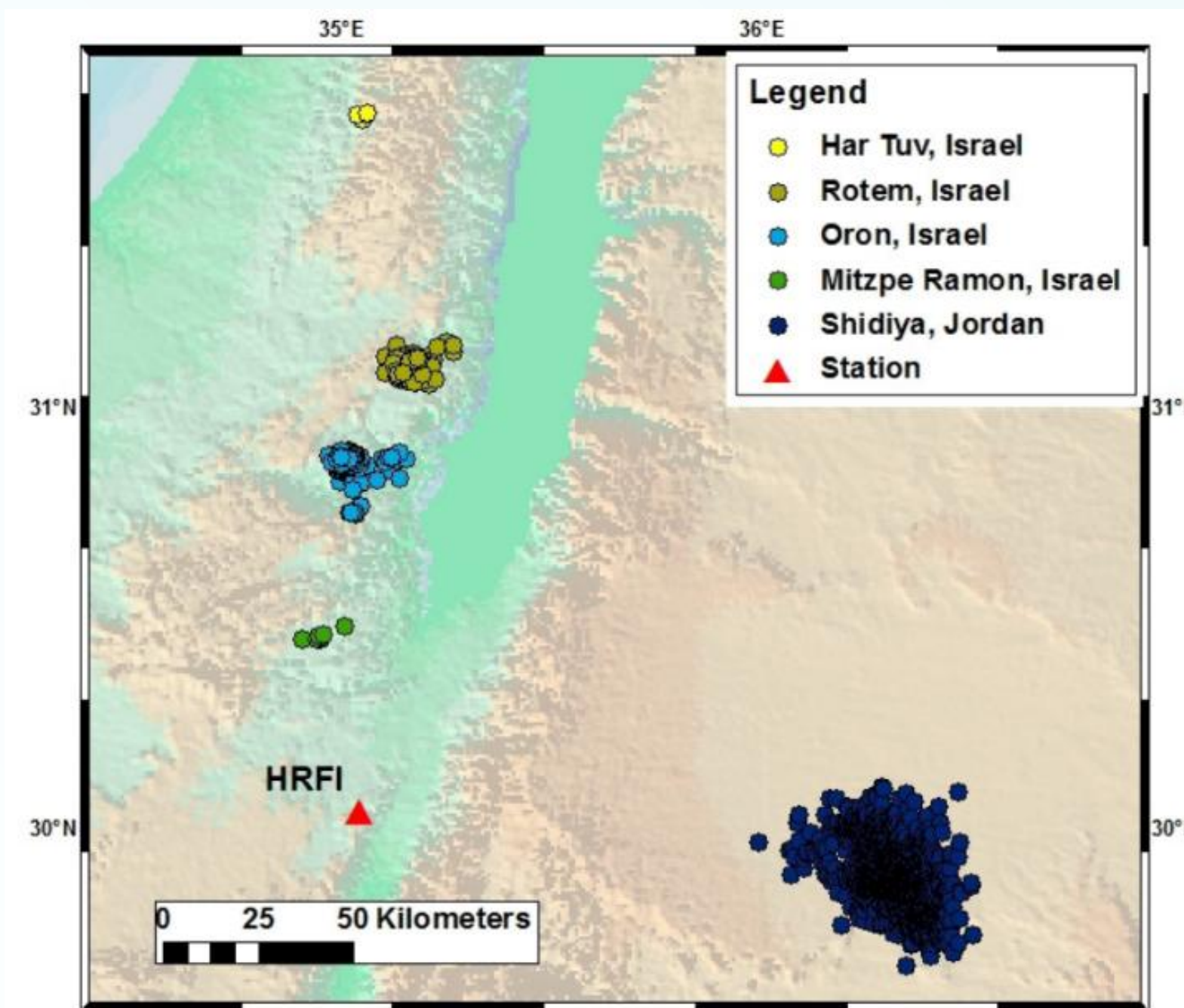


Figure 1: Map of quarry clusters.

Objectives

Our main objective is to develop an open set classifier for quarry blast explosions.

- Seismic signal representation: Find a robust representation that well separates the explosions and achieves good classification results for closed set few shots classification.
- Based on the selected representation, lay the foundation for an open-set classification with rejection algorithm for seismic classification tasks.

Methodology

We used a dataset that was received from Soreq Nuclear Research Center, which contained several hundreds of quarry explosions signals from 6 different quarries. The signals were recorded at HARIF station near Neot Smadar, Israel. Each signal contains three components, one vertical (Z) and two horizontal (North, East). We assumed that the received signals have a fixed source mechanism and are only explosions. For time series representations we explored several convolution based-methods like MiniRocket [1] and TSFresh[2], which are considered state-of-the-art versus a tailored time-frequency representation. The time-frequency representation is a manipulation of a spectrogram that beans the frequencies and normalizes them[3].

We tested the classification capabilities of each of the representations against different sizes of training and testing sets. In addition, we also used a leading classification method HIVE-COTE2 that does not require a different representation for the signal [4,5].

Furthermore, to handle with open set we propose a new method that address with this challenge, creating a well-defined separation that can reliably reject out-of-distribution (OOD) samples. We propose to first detect the samples that are on the boundary of each class. Then synthesize additional samples beyond the boundary and set a more flexible limit of OOD rejection and detection. Last, application of methods such as one-class support vector machine will be fitted to the augmented data. Figure 2 below illustrates the main steps in the proposed model. The left panel shows the known training classes as well as new data from unknown classes. The central panel illustrates how a closed-set classifier would separate the space, leading to misclassification of the new samples that do not belong to the train classes. The left panel illustrates our proposed approach for open-set classification. It includes boundary detection and augmentation followed by an open-set classification model.

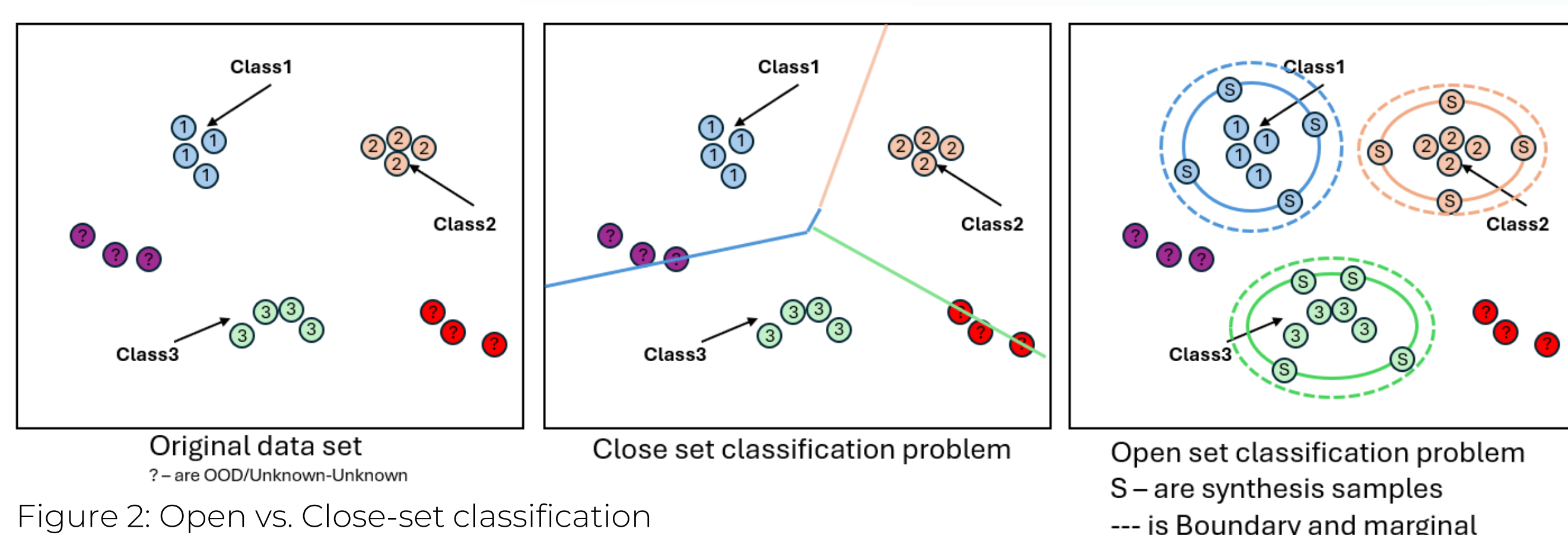


Figure 2: Open vs. Close-set classification

Results

We found that the leading representation for quarry explosion signals is a sonogram. Figure 3 shows a single seismic event with its 3-component input on the left together with the corresponding sonogram representation on the right. The sonogram representation achieved higher results compared to MiniRockets, TSFresh and HIVE-COTE2. The classifier we chose to use is RidgeClassifierCV that achieved 100% accuracy for a (85%-15% train-test split) and for a few-shot training configuration, where the number of training examples for each class is limited, it achieved over 95% accuracy. MiniRocket achieved ~88% accuracy, TSFresh achieved ~74% accuracy and HIVE-COTE2 achieved ~89% accuracy.

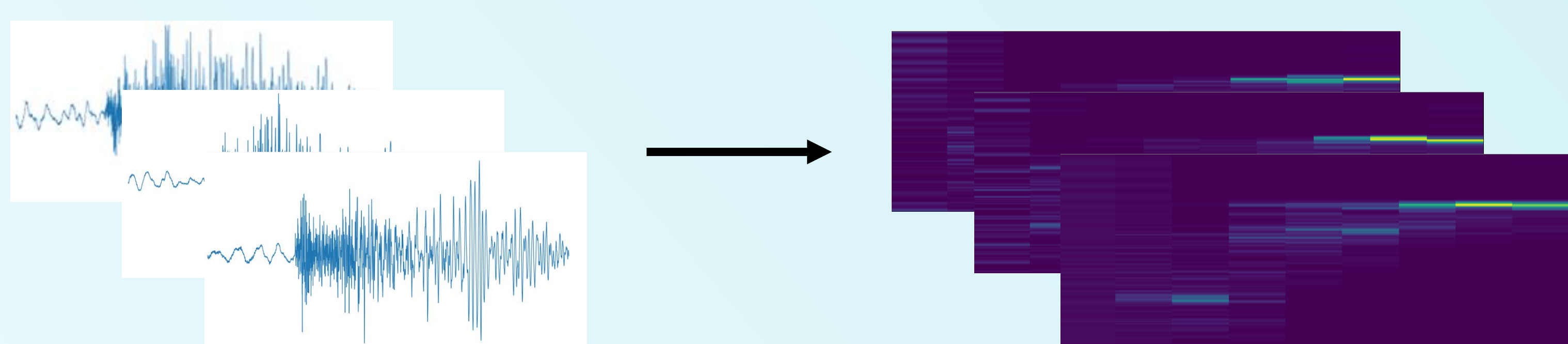


Figure 3: Left: A single 3-channel seismic event. Right: Corresponding Sonogram representation.

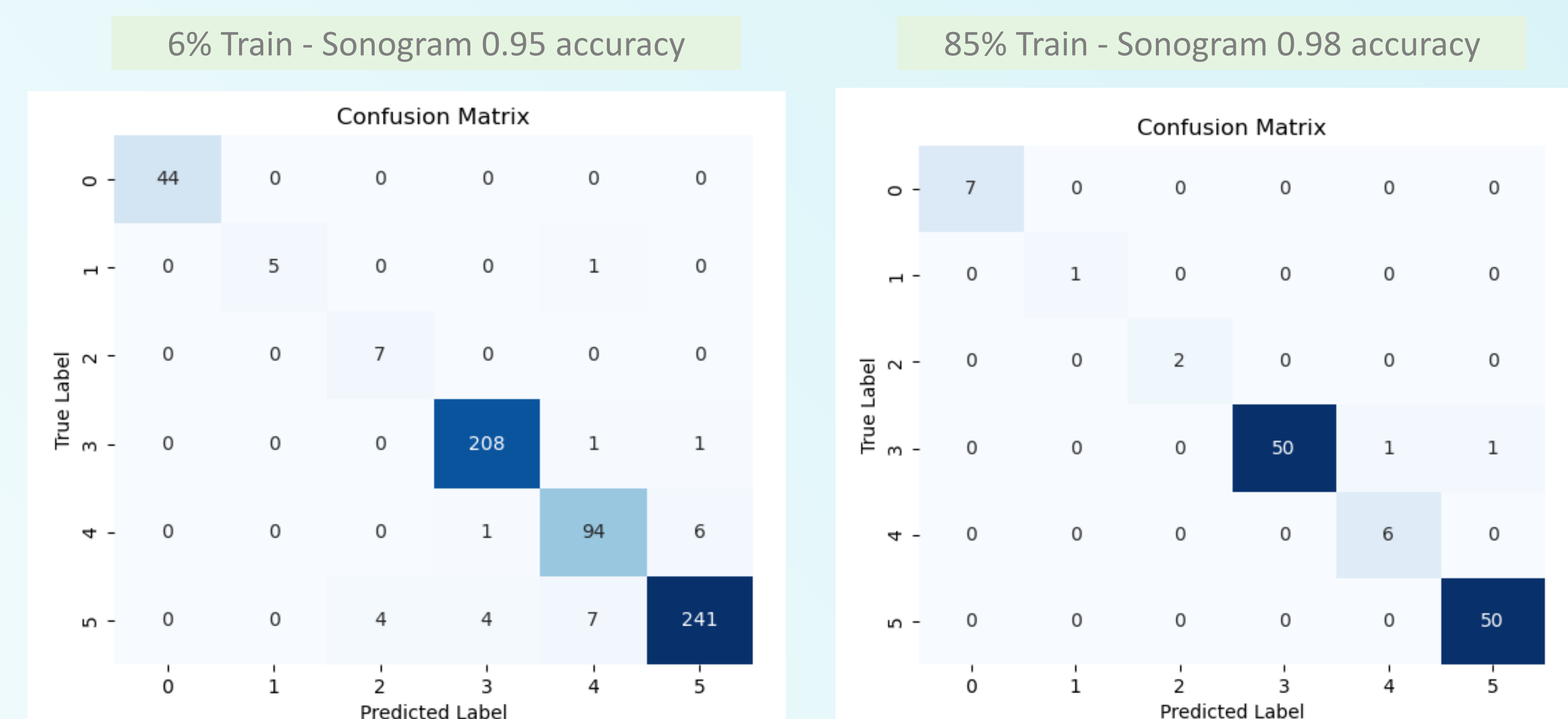


Figure 4 plots the confusion matrices for both cases.

Conclusion

This work deals with seismic signal classification, with a long-term goal of developing an open-set classifier and a short-term goal of comparing between different time series representations methods in the setting of a closed-set classifier. For the first part, we applied and compared 4 different time series representation methods in the setting of a closed-set classifier, once with a typical 85%-15% train-test split and once with a small, few-shot training set. We found that a tailored time-frequency representation named sonograms yields the best results for both settings.

Additionally, we formulated the next steps for open-set classification. These include detection of boundary samples for each class based on the sonogram representation followed by data augmentation near and out from the detected boundary samples. Last, we plan to fit an open-set classification method to the augmented data. We believe that this framework will be useful for geophysical analysis and other time series classification tasks.

References

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